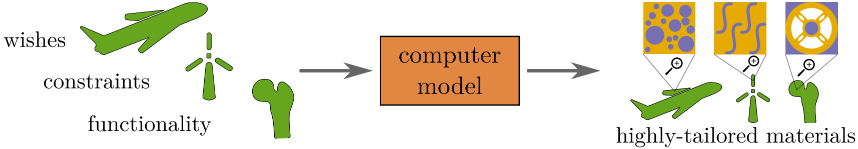
Materials and computer models – avenues for progress

The answer: combining a handful of experiments with extensive virtual testing

Design optimization

- Design variables can be tweaked at different scales (e.g. microstructures, micro properties)
- Finding the optimum among millions of possible combinations ⇒ more efficient designs

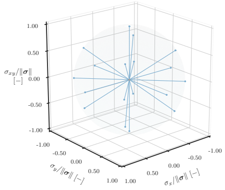


PRNNs versus conventional ML models

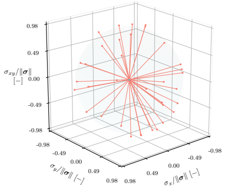
Training and testing with the same type of data:

PRNN (18 monotonic curves) RNN (576 curves with unloading)

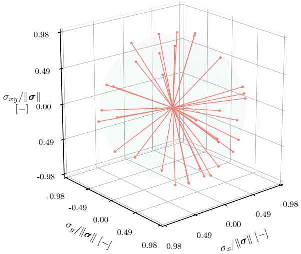
Train

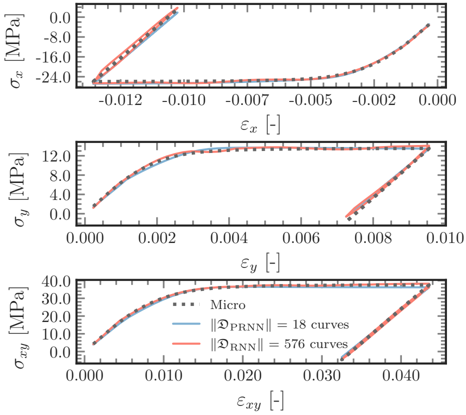


or



Test





[Maia *et al.*(2023), *CMAME*]

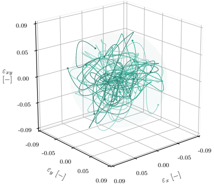
PRNNs versus conventional ML models

Great, but now we can do the simple ones again... right? No!

PRNN (36 curves)

RNN (576 curves)

Train



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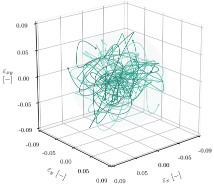
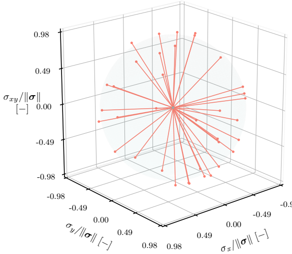
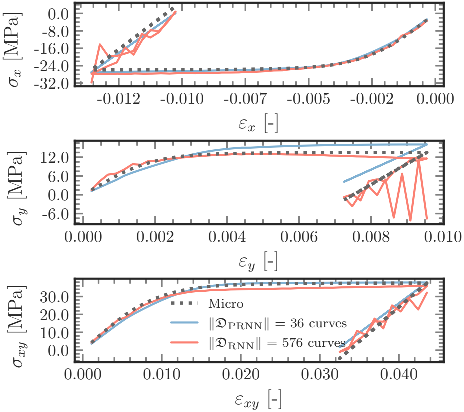


Test





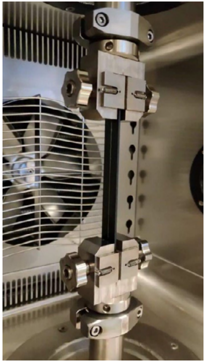
[Maia *et al.*(2023), *CMAME*]

What we are now able to do: FE^2 versus experiments

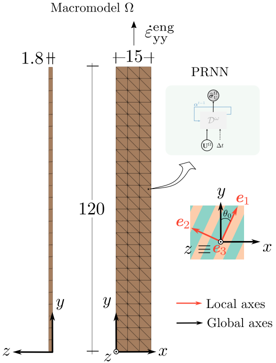
We also used it to confirm some experimental results that had been puzzling us:

- Uniaxial tension on off-axis carbon-PEEK coupons

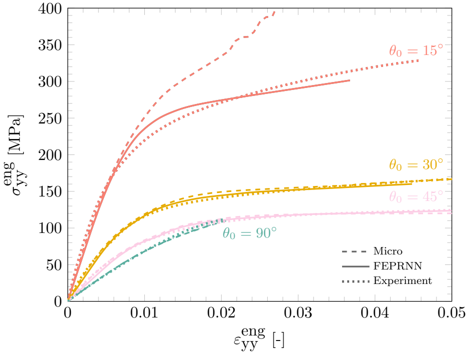
Uniaxial tension experiment



Multiscale model



Engineering stress-strain

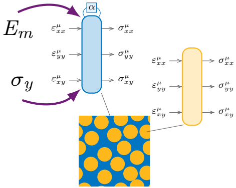


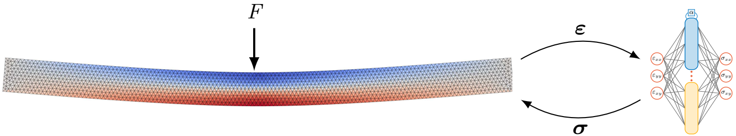
[Maia *et al.*, *(In preparation)*]

What we are now able to do: Multiscale UQ

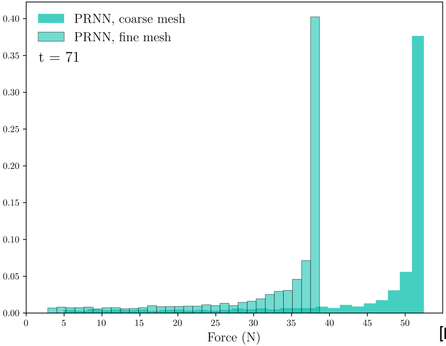
Three-point bending, fine macro mesh:

- Monte Carlo forward UQ, $N = 5000$ samples
- PRNN trained with a single (E_m, σ_y) pair
- Sequential FE^2 UQ would now take $\approx$50 years to run

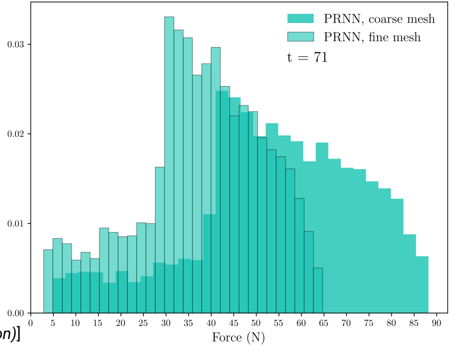




$$p(F) = \int p(F|E_m)p(E_m)\,\mathrm{d}E_m$$



$$p(F) = \int p(F|E_m,\sigma_y)p(E_m,\sigma_y)\,\mathrm{d}E_m\mathrm{d}\sigma_y$$



[Kovacs et al, *(In preparation)*]